

B30801 Managerial Accounting Autumn 2025

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Session #4: Responsibility centers and budgets; Transfer pricing; Variance analysis.

CHICAGO BOOTH



The University of Chicago Booth School of Business

Responsibility Centers and Budgets

Rules of the game

There are three elements of an organization's Rules of the Game:

- Assign **decision rights**.
- Performance measurement and evaluation.
- Reward and punishment system.

Decision rights: the enforcement and disciplinary powers of the top-level executive office will be used to enforce your ability to take an action.

Defining **responsibility centers** and setting **budgets** are key elements of the *Rules of the Game*—they define how decision rights (over acquisition and use of resources) are allocated, how performance is measured, and how business unit managers are evaluated and rewarded.

Tools for assigning decision rights, and then to assess performance.



This is Spruce 2, a 750 MW coal-fired power station operated by CPS Energy in Texas to supply power to San Antonio.

Responsibility Centers

1. **Cost Center.** Manager of Spruce 2 is asked to provide certain level of output (power) depending on demand conditions that are best assessed by CPS Energy, which manages the full grid.
 - Unit is evaluated based on **reliability** and **cost** (per KW hour).
 - Each year, plant manager negotiates with central decision-maker regarding an operating budget which may be a **flexible budget** (i.e., linked to output).
 - Plant manager's goal: **minimize cost**.

Responsibility Centers

- The cost center set-up works well here given: (1) **specific knowledge** needed to operate plant, and (2) **quantity and quality of output** easy to observe and measure.
- Manager is responsible for cost but not revenue decisions.

Works when unit manager has **specific knowledge** about the most efficient use of factor inputs, operating the unit, etc. (i.e., the production process).

Responsibility Centers

Cost centers *don't* work well when **quality is hard to measure**—if quality is hard to measure, managers can lower costs by reducing quality and use these resources to expand their operations and responsibilities – “**empire-building**”.

- IT or HR are common examples (e.g., head of IT reduces quality of services and uses those resources to expand scope of operation).

I might compromise on quality just to meet the targeted budget. example; HR budget is \$500k per year. I will aim to underspend (if quality is unobservable) and then appear to be performing well as a HR manager.

Responsibility Centers

- 2. Revenue Center.** Natural complement to cost center. Goal: **maximize revenues** subject to resources made available to unit. Unit manager empowered to deploy marketing/sales resources to maximize revenues with little or no role in production.
- Booth Advancement. Team of 50 people who fund-raise for the school.
 - Goal is to achieve two annual targets—for cash and FRP (“fund-raising progress”) while maintaining good relations with donors, most of whom are alumni (**quality**).

Responsibility Centers

Manager has budget target. How is this set?

Budget “**ratcheting**”:

- If I achieve my goal, higher target set for next period;
- If I fall short, target unchanged.

→ *If I'm ahead of my goal, I hold back additional sales to avoid target being increased further (“sand-bagging”); no incentive to exceed goal by significant amount. We may want to have discretion to award a higher bonus if targeted exceeded.*

Responsibility Centers

How should we set targets?

- Participation. Involving the unit manager and her team in setting the target increases buy-in and improves performance.
- Set a “stretch” target? As long as someone is committed to a goal, **has the ability to achieve it**, and does not have conflicting goals, there is a *positive relation between goal difficulty and task performance*. [But use of this term – particularly if “top down” – can raise red flags with investors, regulators, etc.]

Responsibility Centers

- 3. Profit Center.** Use when business unit manager has best information (specific/local knowledge) for decisions about marketing and sales, product mix, operations, and production [i.e., drivers of both revenues and costs].
- Business unit manager internalizes cost and revenue effects of her decisions (e.g., expanding sales team).
 - Problems due to “gaming” of reported numbers – definition and measurement of profit, including revenue recognition, indirect cost allocations, etc.

Responsibility Centers

Example; avoid incurring discretionary maintenance expenses etc.

- Usually based on *annual* measures. Problems arise when there is a tradeoff between short and long term performance and manager has short horizon (based on how they're evaluated and compensated).
 - At Spruce, manager who retires next year may reduce maintenance expenditures (expenses) to achieve annual profit target and receive a bonus even though this will lead to higher maintenance costs in long run. Known as the **horizon problem**.

- cannibalisation / inter competition
- wrong / excessive transfer pricing (or fixed allocations) from other business units
- may get complacent if they are excessively attributed for GP that does not come from their effort.

Responsibility Centers

- Profit centers don't work well if there are interdependencies across divisions in either sales or production.
- If products are to some degree substitutes (high-end Toyotas vs. Lexus), managers of business units compete in ways that reduce overall firm value.
- This was the problem with setting up a division to operate the 3D MRI machine in VMD.

Sequoia China's push into Singapore sets up fight against Indian arm

Plans to expand into south-east Asia dealmaking from city-state puts it in direct competition with India-based Peak XV



Sequoia China, rebranded as HongShan, is managed by Neil Shen, who led early investments in Alibaba and TikTok owner ByteDance © Bryan van der Beek/Bloomberg

Mercedes Ruehl in Singapore and **Kaye Wiggins** in Hong Kong JULY 8
2023

Sequoia Capital's China unit is drawing up plans to expand into Singapore, putting it in competition with another arm of the venture capital group as the firm divides into three amid rising geopolitical tensions.

HongShan, the Chinese arm of the Silicon Valley group, has set up an office in the city-state and is working on plans to use it as a base for investing in south-east Asia, multiple people with knowledge of the situation said.

That would pit the Chinese investment juggernaut, which manages \$56bn in assets, against Peak XV Partners, [Sequoia's](#) India and south-east Asia business, which already has an office in Singapore.

Both units, along with Sequoia's US-based business, will [split into separate companies](#) by 2024, the group said last month.

In the *Epic vs. Apple* litigation, Epic accused Apple of running its App Store as a monopoly, and provided an expert who claimed its operating margin was above 70%.

Apple countered that it doesn't calculate a profit margin (or produce an internal management P&L) for the App Store and doesn't manage its business units as profit centers.

Here's Tim Cook's explanation (testimony):

1 A. Yes.

2 Q. Now, let's go back to Apple's 2020 10-K. And I would like
3 to talk about some of the P&L and profitability issues that
4 have been raised with the Court.

5 The 2020 10-K is DX4581.

6 A. 4581.

7 Q. And if we go to page .022 in that document.

8 Do you see information there for total net sales and net
9 income for 2020, Mr. Cook?

10 A. Yes.

11 Q. Can you use that information to determine what Apple's
12 profit margin was for fiscal year 2020?

13 A. Yes. If you divide the 57.4 billion by the 274.5, I
14 believe, if my memory serves me correct, you get to 20.9
15 percent.

16 Q. And, sir, do you consider any other profit margin in
17 running the business?

18 A. It's not the way we do it.

19 Q. And does Apple prepare fully burdened P&L statements for
20 today's business units --

21 A. No.

22 Q. -- in the ordinary course of business?

23 A. No.

24 Q. Can you explain why not?

25 A. Well, because, first of all, allocation of costs, of joint

Fully burdened; means the entire
business's costs are fully allocated

1 costs, are very difficult to do, and it's open for debate

2 about how to do it. And so you would wind up getting the

3 company focused on arguing between the different areas about

4 where costs should be and it would be totally unproductive, in

5 my mind.

6 Steve actually did this when he came back to the company.

7 If I remember correctly, he told me a story once that he found

8 Apple losing a billion dollars a year but all of the divisions

9 all making money. And so you can imagine what he then did.

10 He just -- he blew it up.

11 And I've never wanted to go back to that point. We just

12 always kept it as one.

13 Q. Has the practice of looking at the company's profitability

14 overall enhanced the company's abilities to compete, in your

15 view?

16 A. I think so, because it means that we spend more time

17 focusing on customers and not focusing on each other.

18 Q. Has Apple ever attempted to determine the specific

19 profitability of the App Store as a stand-alone business unit?

20 A. No.

21 Q. Do you believe the App Store is profitable?

22 A. Yes, I do.

23 Q. Are you able to give us a specific number on

24 profitability?

25 A. No. We haven't done that, but, you know, I have a feel,

Source: Tim
Cook testimony,
Epic v. Apple,
5.21.21

Common budget games:

- Spend down budget just before end of period (if cost center budget allocations don't roll over).
- Accelerate revenues (e.g., channel-stuffing; special deals; discounts) to hit profit targets.
- Defer or reduce “discretionary” spend (e.g., R&D, maintenance) to hit profit targets.
- “Big bath”: If going to miss target, miss by a lot – write-off costs now to reduce future expenses and juice metrics next period.
- “Sandbagging”: Once target for period is achieved, hold back further revenues to lower future targets.

$$ROE = \frac{NI}{SE}$$

- net income
- shareholder equity
- EBIT
- tangible assets

$$ROA = \frac{EBIT}{TA}$$

Responsibility Centers

- 4. Investment Center.** Evaluate managers based on return on investment (ROI). Makes sense if unit manager is also best positioned to make *investment decisions*.
- Measures such as return on assets (ROA), return on invested capital (ROIC), return on investment, etc.
 - Helps solve profit center incentive problems (e.g., managers cut maintenance expenditures to hit profit targets—i.e., underinvestment).
 - Once again, metrics are usually short-term/annual and can be gamed depending on accounting rules (e.g., write-offs increase future measured ROI).

A common measure is ROIC:

$$ROIC = \frac{NOPAT}{Invested\ Capital}$$

NOPAT = Net Operating Profit after Tax

(Although may want a pre-tax measure if business unit manager cannot influence effective tax rate)

Responsibility Centers

- Requires measurement of assets (invested capital) at business unit level, which may require allocation of joint assets/capital (if it can be done at all).
- Data centers at Google are a shared resource that support many different product groups.
- **ROI metrics ignore scale.** For example, a new project may be positive NPV but if ROI is below business unit average, manager will not invest.
- This **may result in divisions shrinking in size (under-investment problem)**—would you prefer a 7% return on \$1M or a 5% return on \$5M?

NPV gives the value at the start of the project.
EVA assigns the value to each year that the project is running. Reasonable attempt to showcase that R&D is not expensed upfront (as per normal accounting practice)

Responsibility Centers

5. EVA/Residual Income.* Unit evaluated based on dollars of residual income (**Economic Value Added**) after subtracting cost of capital.

- Solves many accounting measurement and incentive problems of profit and investment centers.
- Residual income = Net income minus cost of capital charge (cost of capital * capital invested).
- Consistent with NPV if implemented with right earnings measure.

*EVA is a measure of performance, not a responsibility center in its own right (could be used for a profit center).

Responsibility Centers

- EVA is a measure of economic profit:

$$\text{Economic Profit } \$ = \text{Invested capital} \times (\text{ROIC} - \text{WACC})$$

ROIC - cost of capital

- Best “flow” measure of performance. Avoids scale problems of ROI. Tells us amount of value created by the business, which is exactly what shareholders care about (that is, is $\text{ROIC} > \text{WACC}$?)
- EVA^{TM} includes adjustments to GAAP earnings to correct its defects as a measure of performance (e.g., capitalize and amortize R&D expenditures). Measure is then tied to evaluation and compensation.

$$EVA_t = C_t + \overbrace{(I_t - I_{t-1})}^{\text{Economic depreciation}} - \overbrace{I_{t-1}r}^{\text{Dollar cost of capital}}$$

C_t = operating cash flow,

I_t = **economic value** of investment capital, end of year t,

r = cost of capital (e.g., from CAPM).

EVA (Residual Income)

Key features:

- Consistent with the NPV rule.
- Adjusts for distortions in accounting as a measure of economic performance (e.g., distortions due to GAAP). For example, may want to capitalize certain R&D costs.
- Imposes a cost of capital charge on managers (accounting earnings subtracts cost of debt only).
- Takes account of *economic* depreciation of capital base.

Budgets and Variance Analysis

Budgets

- **Budget:** Plan for business or business unit **for a specified period** of time, **expressed in financial terms** – a financial **plan** prepared in advance of each period.
- How a budget is specified (in terms of costs, revenues, profit, or ROIC, etc.) depends on how we define responsibility centers
 - this is the mapping from allocation of decision rights to performance measurement and evaluation.

Budgets

- **Assign decision rights:** define resources available and under control of the business unit. Connected to definition of responsibility center.
- **Communicate and coordinate** information about decision rights/resource allocations both horizontally and vertically.
- Set **shared goals** through negotiation (iterative process; top-down and bottom-up; participative process).
- **Measure and evaluate performance.** Evaluated and rewarded relative to performance delivering budget targets.

FIGURE 2: ROLLING FORECASTS AT HILTI

JUNE-SEPTEMBER

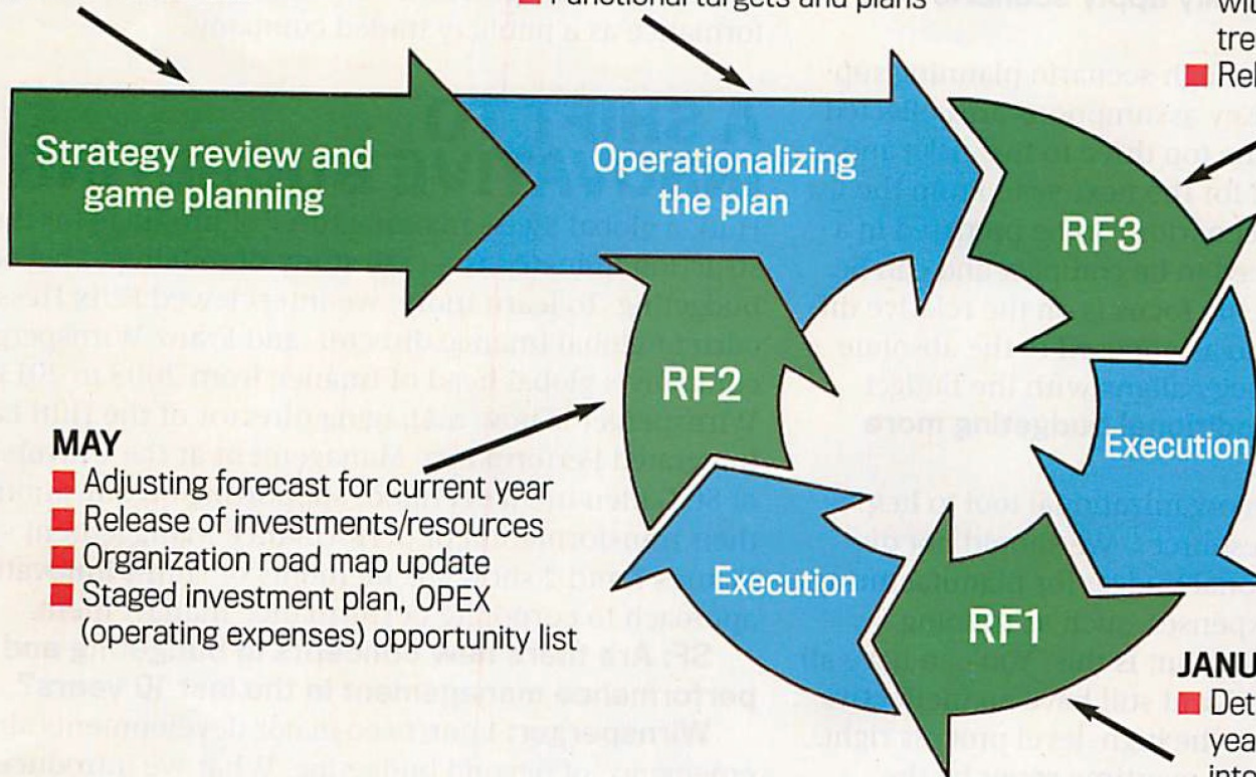
- Strategy implementation review
- Need for action
- Calibrated game plan on financial ambition for next year, including initiatives and resources

SEPTEMBER-DECEMBER

- Operationalizing game plan
- Marketing/Sales: resource planning/territory splitting and target setting, road map
- Compensation models and targets/curves
- Functional targets and plans

OCTOBER

- Precise latest estimate for current year
- First forecast for next year (in line with game plan and confirmed by trends and investments)
- Release of investments/resources



MAY

- Adjusting forecast for current year
- Release of investments/resources
- Organization road map update
- Staged investment plan, OPEX (operating expenses) opportunity list

JANUARY-FEBRUARY

- Detailed bottom-up plan current year; changes in environment or internal trends
- Release of investments/resources

Budgets

Goal mechanisms—what makes them work?

- Goals are **directive**: goals direct attention and effort toward goal-relevant activities and away from irrelevant activities.
- Goals **energize**: high goals lead to greater effort.
- For goals to be effective, people need **feedback** that reports progress towards goals.
- People more likely to buy-in if they **participate** setting the goal.

Zero-based budgeting

- In most scenarios, budgets are adjusted upwards from one year to the next (i.e., they ratchet or increase based on inflation or organizational growth rate).
- In addition, business unit managers request incremental resources for new activities (e.g., the marketing manager needs more headcount for a social media team).
- Zero-based budgeting requires the unit manager to justify the **entire** budget each year, not just the annual increase.

Very common in private equity companies. The invested company management will have to justify almost every line item (as an effective way for reducing costs)

WE'RE DOING "ZERO-BASED BUDGETING" NOW, SO
WE HAVE TO JUSTIFY EVERY EXPENSE EACH PERIOD.

UNFORTUNATELY, WE DIDN'T GET APPROVAL TO TURN
ON THE LIGHTS, BUT WE'LL TRY AGAIN NEXT QUARTER.



TOM
FISH
BURNE

Variance Analysis:

Why did we miss our planned profit target?

- Why did our sales miss goal?
- Why did our products cost more than budgeted?
- Who's responsible?

Key tool for performance *evaluation*.

Variance = Actual – Budgeted

- Applies to both revenues and costs.
- **Favorable** variance (higher operating income):
Revenues > Budgeted
Costs < Budgeted
- **Unfavorable** variance (lower operating income):
Revenues < Budgeted
Costs > Budgeted

P = Actual Price (or Cost)

Q = Actual Quantity

EP = Budgeted Price (or Cost)

EQ = Budgeted Quantity

Total variance = $P.Q - (EP.EQ)$

= $[Q.(P - EP)] + [EP.(Q - EQ)]$

= Price variance + Quantity variance

This applies to both revenues and costs

Why is this useful?

- Helps us **understand** and analyze **why** actual performance differs from budget (or goal) set for manager/business unit.

Why is this useful?

- Helps us **understand** and analyze **why actual performance differs from budget** (or goal) set for manager/business unit.
- Allows us to **assign** underlying drivers of performance **to different managers based on responsibilities**. For example:
 - Actual production volumes depend on demand, which the plant manager cannot control.
 - Pricing decisions may not be under the control of the sales team. Mix decisions may reflect changes in customer tastes that are unrelated to marketing effort.

Transfer Pricing

What is a transfer price?

Price at which goods and services are transferred between business units **within** organizations. Can set at:

- Variable cost (of transferor).
- Full cost (of transferor).
- Full cost + profit margin (of transferor).
- External market price.
- Internal price negotiated by business units.
- Dual price (exit price for transferor is different from entry price for transferee).

What is a transfer price?

- The way transfer prices are set is part of the organization's "rules of the game." Transfer prices will be most important when profit centers are used – transfer prices determine revenues (COGS) for the selling (buying) unit.
- Transfer prices are for internal transactions and do not affect the entity's overall economics.
- As with consolidated financial statements, only transactions with entities outside the organization have **direct** economic effects for the entity as a whole.

What is a transfer price?

- However, transfer prices affect unit managers' decisions about transacting with external entities and therefore have **indirect** economic effects on overall entity.
- For multinationals, transfer prices affect the way the entity as a whole is taxed if different units are in different tax jurisdictions with different tax rates. But this is **completely separate** from how the organization is managed internally.

Transfer pricing and taxes

- Consider a drug-maker domiciled in Ireland ($\tau = 0.15$) that licenses a U.S. affiliate ($\tau = 0.35$) to manufacture, market, and sell one of its drugs in the U.S.
- In 2024, sales of the drug to U.S. customers are \$100M, with the U.S. affiliate incurring \$20M in direct variable costs to manufacture and market the drug.
- The Irish parent receives royalties from the U.S. affiliate in exchange for the right to use its IP to manufacture and distribute the drug product. The royalties are a form of transfer price.

	Irish Parent	US affiliate	Consolidated
τ	0.15	0.35	
Revenues	\$50M*	\$100M	\$100M
Variable direct costs	-	(\$20M)	(\$20M)
Royalties	-	(\$50M*)	-
Taxable income	\$50M	\$30M	\$80M
Tax	\$7.5M	\$10.5M	\$18M
ETR			22.5%
* Denotes transfer price based on valuation of technology.			

	Irish Parent	US affiliate	Consolidated
τ	0.15	0.35	
Revenues	\$50M*	\$100M	\$100M
Variable direct costs	-	(\$20M)	(\$20M)
Royalties	-	(\$50M*)	-
Taxable income	\$50M	\$30M	\$80M
Tax	\$7.5M	\$10.5M	\$18M
ETR			22.5%

Let's increase the transfer price to \$80M, shifting earnings from the US affiliate to the Irish entity.

	Irish Parent	US affiliate	Consolidated
τ	0.15	0.35	
Revenues	\$80M*	\$100M	\$100M
Variable direct costs	-	(\$20M)	(\$20M)
Royalties	-	(\$80M*)	-
Taxable income	\$80M	0	\$80M
Tax	\$12M		\$12M
ETR			15%

Earnings are shifted – via the transfer price – to the Irish parent, lowering the overall tax rate (and taxes paid to US).

Transfer pricing and taxes

- According to the US tax code and regulations (S482), transfer prices should be determined based on *arm's length pricing* based on: (a) valuation of intangible assets (intellectual property), and (b) cost allocations.
- The facts in my example are loosely based on the GlaxoSmithKline case.
- You will not see any tax-based examples on the exam.

EVA example*

*Supplementary and non-examinable content.

EVA Example.

- Initial capital investment \$200, depreciated for book (and tax) purposes straight-line over five-year life to zero salvage.
- Tax rate is 35%; cost of capital is 10%.
- Project generates operating cash flows (EBITDA) of \$100 per year for five years.
- Let's first compute NPV.

	0	1	2	3	4	5
NPV Computation						
EBITDA		100	100	100	100	100
Depreciation		(40)	(40)	(40)	(40)	(40)
EBT		60	60	60	60	60
tax @ .35		(21)	(21)	(21)	(21)	(21)
Earnings		39	39	39	39	39
add back Non-cash expense		40	40	40	40	40
minus Capex	(200)	0	0	0	0	0
FCF	(200)	79	79	79	79	79
 PV factor	 1	 0.9091	 0.8264	 0.7513	 0.6830	 0.6209
PV	-200.00	71.82	65.29	59.35	53.96	49.05
NPV	99.47					

EVA computations

- To compute EVA, we need to compute both economic depreciation of the capital asset as well as the cost of capital charge.
- To get the total amount, we first compute the annual *total* cost of capital charge, which is that annual five-year cash flow that yields a PV of \$200 with a 10% cost of capital.
- This amount is \$52.76 (an annuity).

	0	1	2	3	4	5
End of year asset value	200	167.24	131.21	91.57	47.96	0
Economic depreciation		32.76	36.03	39.64	43.61	47.97
Cost of capital @ 10%		20.00	16.73	13.12	9.15	4.79

Each year, asset value is the PV of the remaining cash flows (5-year annuity of \$52.76 discounted at 10%).

-- \$200 is the PV of an annuity of \$52.76 for five years at 10%; \$167.24 is the PV of an annuity of \$52.76 for four years at 10%; etc.

Economic depreciation is the change in the asset value (the sum of the economic depreciation over the asset's life is \$200).

Cost of capital is 10% of opening asset value.

	0	1	2	3	4	5
EVA Computation						
EBITDA		100	100	100	100	100
tax (as per above)		(21)	(21)	(21)	(21)	(21)
		79	79	79	79	79
Economic depreciation		(32.76)	(36.03)	(39.64)	(43.61)	(47.97)
Cost of capital charge		(20.00)	(16.73)	(13.21)	(9.15)	(4.79)
EVA		26	26	26	26	26
 PV factor		0.9091	0.8264	0.7513	0.6830	0.6209
PV		23.85	21.69	19.65	17.92	16.29
Total PV of EVA	99.40					

Note that the PV here is the same as the NPV number derived above. Essentially what we have done is taken the initial \$200 investment and allocated it to each year in an economically meaningful way that provides an annual measure of performance that can be used to evaluate managers (especially useful for comparing managers across different units of the organization). Note that this cannot be done with the NPV computation.